

A Partial Answer for Ball Families in the James Function Space Problem

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Status. Partial result, likely valid. The packet proves the nonseparability-of-the-dual part of Problem 1 in arXiv:1210.2379 for the Euclidean-ball and $\ell_\infty^{d_0}$ -ball families. It does not prove the remaining assertion that the corresponding spaces do not contain ℓ_1 .

1 Source Problem

The source paper is S. A. Argyros, A. Manoussakis, and M. Petrakis, *Function spaces not containing ℓ_1* , arXiv:1210.2379 [1]. In Section “Remarks and Problems,” Problem 1 asks the following. Let Ω be an open bounded subset of $\mathbb{R}^{d_0}$, $d_0 > 1$, and for a family $Q(\Omega)$ of convex bodies contained in Ω define

$$\|f\|_{\text{JF}(Q(\Omega))} = \sup \left\{ \left(\sum_{i=1}^n \left(\int_{V_i} f \right)^2 \right)^{1/2} : \{V_i\}_{i=1}^n \subset Q(\Omega), V_i \cap V_j = \emptyset \right\}.$$

The problem asks for which families $Q(\Omega)$ the following hold:

- (i) $\text{JF}(Q(\Omega))$ does not contain ℓ_1 , (ii) $\text{JF}^*(Q(\Omega))$ is nonseparable.

The paper notes that this was open, in particular, when $Q(\Omega)$ is the family of Euclidean balls or the family of $\ell_\infty^{d_0}$ balls contained in Ω .

2 Proof Intuition

The dual-nonseparability half has a simple mechanism that does not require the delicate weak-Cauchy analysis used in the paper for excluding ℓ_1 . Every set V in the defining family gives an integration functional $V^*(f) = \int_V f$, and normalized indicators provide exact norm tests for these functionals. If two sets U, V differ by positive measure, then the normalized indicator of $U \setminus V$ (or $V \setminus U$) separates U^* from V^* by distance at least one. Thus any uncountable family of sets that are pairwise distinct modulo null sets yields an uncountable 1-separated subset of the dual.

For Euclidean balls and ℓ_∞ balls, such an uncountable family is obtained by fixing a center in Ω and varying the radius through an interval small enough that all the balls remain in Ω . This proves part (ii) for the ball families singled out in the source problem. The argument does not address part (i), the absence of ℓ_1 .

Problem 1. Let Ω be an open bounded subset of $\mathbb{R}^{d_0}$, $d_0 > 1$. For $Q(\Omega)$ a family of convex bodies contained in Ω we consider the space $JF(Q(\Omega))$ endowed with the following norm

$$\|f\|_{JF(Q(\Omega))} = \sup \left\{ \left(\sum_{i=1}^n \left(\int_{V_i} f \right)^2 \right)^{1/2} : \{V_i\}_{i=1}^n \subset (Q(\Omega)), V_i \cap V_j = \emptyset \right\}.$$

For what families $Q(\Omega)$ the following hold:

- (i) The space $JF(Q(\Omega))$ does not contains ℓ_1 .
- (ii) The space $JF^*(Q(\Omega))$ is non-separable.

Remark. Our proof for the family of the paralellepeds $\mathcal{P}(\Omega)$ does not yield similar results for other families. In particular the above problem is open if $Q(\Omega)$ is either the Euclidean balls or the $\ell_\infty^{d_0}$ balls contained in Ω .

Figure 1: Problem 1 and the ball-family remark in arXiv:1210.2379, PDF page 45.

3 Statement

Theorem 1. Let Ω be a bounded open subset of $\mathbb{R}^d$ with Lebesgue measure μ , and let Q be a family of positive-measure measurable subsets of Ω . Assume that the displayed formula above defines the norm used to form $JF(Q)$ from $L^1(\Omega)$, and for $V \in Q$ let $V^*(f) = \int_V f$ on the dense copy of $L^1(\Omega)$ in $JF(Q)$.

If Q contains an uncountable subfamily $\mathcal{U}$ such that any two distinct $U, V \in \mathcal{U}$ differ modulo null sets, then $JF(Q)^*$ is nonseparable. More precisely, $\{V^* : V \in \mathcal{U}\}$ is an uncountable 1-separated subset of $JF(Q)^*$.

Corollary 1. Let Ω be a nonempty bounded open subset of $\mathbb{R}^d$, $d > 1$. If $Q(\Omega)$ is either the family of Euclidean balls contained in Ω or the family of ℓ_∞^d balls contained in Ω , then $JF^*(Q(\Omega))$ is nonseparable.

4 Proof

Lemma 1. Let $A \subset \Omega$ be measurable with $0 < \mu(A) < \infty$, and put $g_A = \mathbf{1}_A / \mu(A)$. Then $\|g_A\|_{JF(Q)} \leq 1$.

Proof. For every finite pairwise disjoint family $V_1, \dots, V_n$ from Q ,

$$\left(\sum_{i=1}^n \left| \int_{V_i} g_A \right|^2 \right)^{1/2} = \left(\sum_{i=1}^n \left(\frac{\mu(V_i \cap A)}{\mu(A)} \right)^2 \right)^{1/2}.$$

All terms are nonnegative, so this is at most

$$\sum_{i=1}^n \frac{\mu(V_i \cap A)}{\mu(A)} \leq \frac{\mu(A)}{\mu(A)} = 1,$$

because the sets $V_i \cap A$ are pairwise disjoint subsets of A . Taking the supremum over all such families gives the claim. $\square$

Proof of Theorem 1. For each $V \in Q$, the functional V^* is bounded with norm at most one, since the singleton family $\{V\}$ is permitted in the definition of the norm:

$$\left| \int_V f \right| \leq \|f\|_{\text{JF}(Q)}.$$

The bound is sharp. By Lemma 1, $\|\mathbf{1}_V/\mu(V)\|_{\text{JF}(Q)} \leq 1$, while

$$V^*(\mathbf{1}_V/\mu(V)) = 1.$$

Thus $\|V^*\| = 1$.

Now let $U, V \in \mathcal{U}$ be distinct. Since they are distinct modulo null sets, either $\mu(U \setminus V) > 0$ or $\mu(V \setminus U) > 0$. In the first case set $A = U \setminus V$; in the second set $A = V \setminus U$. Suppose for definiteness that $\mu(U \setminus V) > 0$. Lemma 1 gives $\|g_A\|_{\text{JF}(Q)} \leq 1$, and

$$(U^* - V^*)(g_A) = \int_U g_A - \int_V g_A = 1 - 0 = 1.$$

Hence $\|U^* - V^*\| \geq 1$. The other case is identical after reversing the roles of U and V . Therefore $\{V^* : V \in \mathcal{U}\}$ is uncountable and 1-separated. A separable metric space cannot contain an uncountable 1-separated subset, so $\text{JF}(Q)^*$ is nonseparable. $\square$

Proof of Corollary 1. Choose $x_0 \in \Omega$ and $R > 0$ such that the closed Euclidean ball $\overline{B}_2(x_0, R)$ is contained in Ω .

For Euclidean balls, the family

$$\{\overline{B}_2(x_0, r) : 0 < r < R\}$$

has cardinality continuum, and two balls with distinct radii differ by a set of positive Lebesgue measure. Theorem 1 applies.

For ℓ_∞^d balls, choose $0 < r < R/\sqrt{d}$ and use the closed cubes

$$\overline{B}_\infty(x_0, r) = x_0 + [-r, r]^d.$$

The family $\{\overline{B}_\infty(x_0, r) : 0 < r < R/\sqrt{d}\}$ is again uncountable and pairwise distinct modulo null sets, and each cube is contained in $\overline{B}_2(x_0, R) \subset \Omega$. Theorem 1 gives nonseparability of the dual.

The same conclusion holds if the convention in $Q(\Omega)$ uses open balls instead of closed convex bodies, since the boundary has Lebesgue measure zero and the mod-null separation argument is unchanged. $\square$

5 Verification and Limitations

The proof is a direct estimate from the defining norm and uses no computational or external structural theorem. The core inequality is the elementary estimate

$$\left(\sum_i a_i^2 \right)^{1/2} \leq \sum_i a_i$$

for nonnegative a_i , applied to the measures of the disjoint intersections $V_i \cap A$.

The result is limited to Problem 1(ii), nonseparability of $\text{JF}^*(Q(\Omega))$. It does not establish Problem 1(i), namely that the corresponding space $\text{JF}(Q(\Omega))$ does not contain ℓ_1 .

Cheap run-index searches for arXiv:1210.2379 and the core terms “Function spaces not containing ℓ_1 ,” “Euclidean balls,” and “ ℓ_∞ balls” did not find an existing packet for this target. Bounded web searches for the same phrases and arXiv id found no obvious later paper explicitly resolving the ball-family version of Problem 1. This is therefore promoted as a partial theorem for human review, not as a claimed full solution of the source problem.

6 Human Review Recommendation

Reviewers should check that the intended definition of $Q(\Omega)$ in the source paper permits the chosen ball convention and that the completion of $L^1(\Omega)$ under the displayed norm is the intended ambient space. The separation estimate itself is elementary; the main scope issue is that the argument gives no information about the ℓ_1 -noncontainment half.

References

- [1] S. A. Argyros, A. Manoussakis, and M. Petrakis, *Function spaces not containing ℓ_1* , arXiv:1210.2379, 2012.